

Study on Extend Kalman Filter at low speed in Sensorless PMSM Drives

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Abstract — The application of an Extended Kalman Filter(EKF) as nonlinear speed and position observer in PMSM drives has been a mature research project. However, one significant problem still unsolved is the weakly performance at low speed of PMSM drive system that yield to the best estimation performance on high speed range(>5Hz). This paper proposes a novel method — Adjustable DC bus voltage, for low speed EKF sensorless control of permanent magnet synchronous motor (PMSM)drives. The findings from the experiment that the better performance at low speed (>2Hz) by adjusting the DC Bus voltage according to the system reference speed are also presented. Simulation and Experimental results are given to confirm the effectiveness of the proposed method.

Key Words: PMSM; Sensorless control; Extended Kalman Filter; DC bus voltage; Performance at low speed.

I. INTRODUCTION

Permanent magnet synchronous machine (PMSM), which possesses such advantages as high efficiency, high torque ratio, high power density, has been widely used. The control of PMSM requires precise position of rotors and signal of speed to achieve the field Orientation. Traditional movement control system used revolver or encoder to test the position and speed of the rotor. However, not only would these sensors, connectors, and cables increase the cost, but also decrease the reliability of the system and even constrain the adaptability of the system to certain conditions. Thus, it is rather necessary to develop a new observer which has real-time estimation of the position and speed of rotors so as to eliminate this equipment to strengthen the reliability of the system and reduce the cost. In the past few years, lots of researches and papers offered ways to eliminate the sensors, such as Luenberger, Kalman, Nonlinear, MRAS, Redundant etc^[1-5].

The Extended Kalman Filter (EKF) is an optimal estimator in the least-square sense for estimating the states of dynamic non-linear systems, and it is thus a viable and computationally efficient candidate for the on-line determination of rotor position and speed of a Permanent Magnet Synchronous Motor (PMSM). Theoretical basis and digital implementation of EKF have been deeply investigated. However, at least one major drawback of the EKF application to sensorless

drives is still unsolved, that is the poor performance in low speed (<5Hz).

In addition to the influence of estimation algorithm, the precision of system error estimation is mainly determined by the system observability. Voltage and current are the observation of EKF filter. There isn't any impact on Current data in low speed. But, the harmonic component has a high ratio in voltage data. The estimation fluctuates severely because of the slowly changed system error information disturbed by quickly changed random error. So, the higher ratio of signal-to-noise (SNR) of voltage, the better performance would be in low speed. A simple method for high SNR is to reduce the voltage level of DC bus. The validity of the proposed EKF modified method has been tested by experiments on laboratory prototypes.

II. MATHEMATIC MODELING OF THE PMSM

The motor considered in this study is a three phase Permanent Magnet Synchronous Motor. The dynamic model of a symmetrical two poles motor can be described by a set of differential equations where the variables are related to the stator currents, voltages, rotor speed and rotor position. A synchronous (rotor-fixed— dq)reference frame is commonly used for this modeling. The stator voltage equations can be expressed in state space form as follows^[1]:

$$\frac{di_d}{dt} = \frac{1}{L_d}(u_d - R_s i_d + \omega_r \Psi_q) \quad (1)$$

$$\frac{di_q}{dt} = \frac{1}{L_q}(u_q - R_s i_q - \omega_r \Psi_d) \quad (2)$$

$$\Psi_d = L_d i_d + \Psi_f \quad (3)$$

$$\Psi_q = L_q i_q \quad (4)$$

The expression for the electromagnetic torque is:

$$\begin{aligned} T_e &= \frac{3}{2} p (\Psi_d i_q - \Psi_q i_d) \\ &= \frac{3}{2} p (\Psi_f i_q + (L_d - L_q) i_d i_q) \end{aligned} \quad (5)$$

The equations for the mechanical dynamics are:

$$J \frac{d\omega_r}{dt} = T_e - T_L - B_\Omega \omega_r \quad (6)$$

$$\frac{d\theta_r}{dt} = \omega_r \quad (7)$$

Given J is larger enough, equation (6) can be described by the following equation:

$$\frac{d\omega_r}{dt} = 0 \quad (8)$$

III. EXTENDED KALMAN FILTER

The use of an EKF as nonlinear speed and position observer for PMSM drives is a mature research topic. So, this paper will not discuss the formula derivation but set its emphasis on the improvement performance of low speed. According to the reference^[6-7], the EKF can be described by the following equations:

$$\tilde{x}(k+1) = A' \hat{x}(k) + B' u(k) \quad (9)$$

$$\tilde{p}(k+1) = G(k+1) \hat{p}(k) G^T(k+1) + Q \quad (10)$$

$$K(k+1) = \tilde{p}(k+1) H^T(k+1) [H(k+1) \quad (11)$$

$$* \tilde{p}(k+1) H^T(k+1) + R]^{-1} \\ \hat{x}(k+1) = \tilde{x}(k+1) + K(k+1) \quad (12)$$

$$* [y(k+1) - \tilde{y}(k+1)] \\ \hat{p}(k+1) = \tilde{p}(k+1) - K(k+1) \quad (13)$$

$$* H(k+1) \tilde{p}(k+1)$$

For the Permanent Magnet Synchronous Motor, the well established d-q or Park's model is usually used to study its performance. In the synchronous reference frame, the mathematical model used portrays both the electrical and the mechanical behavior of the motor. Choosing as state variables the currents components is i_d, i_q , the rotor speed ω_r and the rotor position θ_r , the PMSM dynamics can be described by the following discrete-time equations:

$$\vec{X}(k+1) = A' \vec{X}(k) + B' \vec{u}(k) \quad (14)$$

$$\vec{Y}(k) = C' \vec{X}(k) \quad (15)$$

Where,

$$\vec{X} = [i_d \quad i_q \quad \omega_r \quad \theta_r]^T, \quad \vec{u} = [u_\alpha \quad u_\beta]^T, \\ \vec{Y} = [i_\alpha \quad i_\beta]^T$$

T — sample time.

It is a common practice to assume the covariance matrices Q, R and P_0 to be diagonal^[8], for the lack of sufficient statistical information to evaluate their off

diagonal terms. On the other hand, practice has revealed that even starting with non-zero off-diagonal values; at steady state off-diagonal terms remains several times smaller than the corresponding diagonal terms. It can be realized that both Q and R depend on the drive parameters, the sampling time, the measurements amplitude and some other secondary factors.

It is found that, the estimated speed always delays the actual speed. This delay depends on the model covariance matrix Q , it is clear that the lower elements $q_{11} = q_{22}$, the lower the delay of the estimated speed in comparison with the effective one. Opposite to q_{11} and q_{22} , the effects of covariance elements q_{33} and q_{44} , are more evident in the EKF closed loop system.

From these considerations it follows that the Q matrix elements that assure filter convergence is:

$$Q = \begin{bmatrix} 4 & & & \\ & 4 & & \\ & & 400 & \\ & & & 0.1 \end{bmatrix} \quad (16)$$

Matrix R concerns the measurements noise. For the system used in the experiments, it yields a matrix R given by simulation.

$$R = \begin{bmatrix} 0.5 & \\ & 0.5 \end{bmatrix} \quad (17)$$

The initial guess for P_0 slightly influences the EKF behavior. The following matrix has been used in this work:

$$P_0 = \begin{bmatrix} 3.43 & & & \\ & 3.43 & & \\ & & 66 & \\ & & & 10 \end{bmatrix} \quad (18)$$

IV. STRUCTURE OF THE CONTROL SYSTEM

The principle of controlling the Permanent Magnet synchronous motor is known as the Vector Control, where the rotor position serves as the angle of reference. This method is analogous to the control of a separately excited DC machine by field and armature currents. According to this principle, the stator currents are transformed into a reference frame of which is moving with the rotor flux. Therefore, a direct and a quadrature component of the stator current vector can be defined which serve as non-interacting components for controlling the magnitude of the flux and the torque respectively. The structure of the PMSM sensorless

drive proposed in this paper is shown in Figure1. Both speed and current loops have a control cycle $T=200 \mu s$, as it is a satisfactory trade-off between algorithm efficiency and microprocessor load factor.

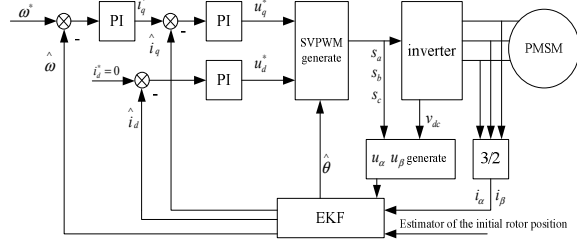


Fig 1. Block diagram of PMSM drive system

V. ANALYSIS OF THE RATIO SIGNAL TO NOISE

There are two main factors^[9] impacting the ratio of voltage signal-to-noise in EKF sensorless control system. One is the measurement error; the other is the influence of the nonlinear inverter. The inaccuracy of the measurement is hardly to make any further improvement in the current condition. However, the influence of the nonlinear inverter can be decreased by adjusting the DC bus voltage.

According to the voltage space vector modulation (SVPWM) principle, suppose the system stays in steady state, that is to say U_{ref} keeps unchanging in harmonic calculation, regardless of the delay of switching on/off as well as the influence of the dead time and the influence of the minimum pulse width, by analyzing the voltage U_{a0} with Fourier decomposition, the harmonic amplitude formula can be presented as the following^[10]:

$$a_n = \frac{2U_{dc}}{n\pi} \left[\sum_{k=1}^{m_f/2} \sin(n\omega_1 t_{2k-1}) - \sum_{k=2}^{1+m_f/2} \sin(n\omega_1 t_{2k-2}) \right] \quad (19)$$

where, $\omega_1 = 2\pi * f_s$, Fundamental angular speed, $m_f = f_c / f_s$, Carrier to fundamental ratio, f_c – Carrier frequency, U_{dc} – Dc bus voltage, U_{ref} – Inverter voltage reference.

As can be seen from the formula (16), the amplitude of each harmonic is in direct proportion with dc bus voltage; therefore, reducing the voltage of dc bus can decrease the amplitude of every harmonic wave and achieve the rise of voltage ratio of signal-to-noise. The following simple stimulation experiment tests the validity of this method.

With the help of the powerful computing function of Matlab/Simulink, The test will be operated by virtual of the software with equivalence output voltage (combined by switched signal and DC bus voltage) and different DC bus voltage level. The reference voltage is sine wave on the stator reference frame :

$$u_\alpha = 20 \cos(\omega_1 t), \quad u_\beta = 20 \sin(\omega_1 t) \\ \omega_1 = 2\pi * 2 \text{ rad/s}$$

Test 1, the DC bus voltage level is 300 volt. Figure 2 is the output voltage and harmonic curve. It is obvious that the total harmonic distortion (THD) is 315.2% derived by Fast Fourier Transform Algorithm from two period's wave starting from 0.5 second.

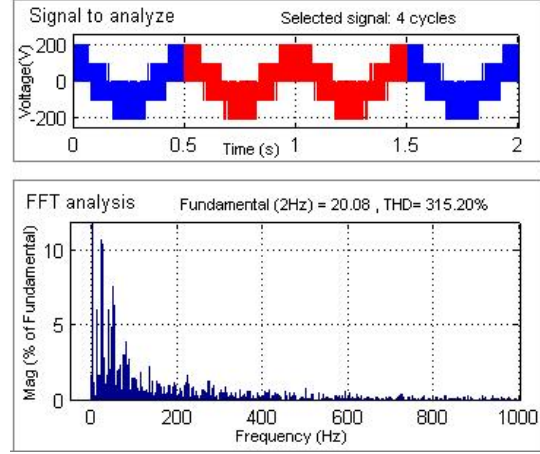


Fig 2. Analysis of the output voltage

Test 2, the DC bus voltage level is 60 volt. Figure 3 is the output voltage and harmonic curve. It is clear that the total harmonic distortion (THD) is smaller 107.38% derived by Fast Fourier Transform Algorithm from two period's wave starting from 0.5 second.

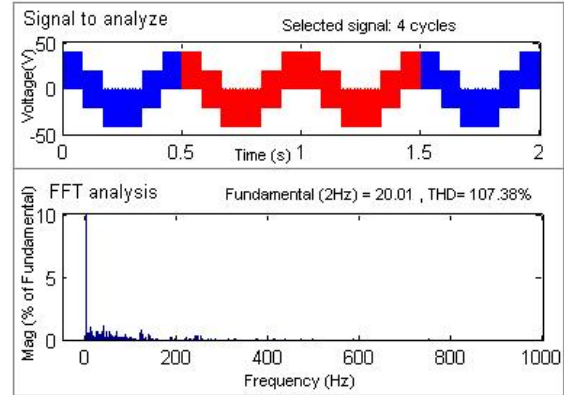


Fig 3. Analysis of the output voltage

From above experimental analysis, it will be concluded that the higher DC bus voltage level, the higher THD will be and then makes a lower SNR. So, a simple and feasible method to get higher SNR is to reduce the DC bus voltage level.

VI. SIMULATION TEST

A. Simulation Structure

The complete drive system is composed of the following parts: the motor, the inverter, the estimator and the controller which is simulated in the computer by MATLAB. The complete system operates in closed loop, i.e., the estimated position generated by the EKF is used as a feedback signal for the controller to drive

the motor.

The motor used in the simulation and experiment is a 1.5Kw interior Permanent Magnet Synchronous Motor. An incremental coder is mounted internally as part of the overall motor construction. The motor parameters as provided by the manufacturer are listed in TABLE 1.

TABLE 1. PMSM PARAMETERS

Quantities	Value	Unit
Rated power	1.5	Kw
Rated speed	1500	rpm
Rated voltage	190	V
Rated torque	9.55	Nm
Rated current	6.1	A
Pole pairs	2	
Phase resistance	1.64	Ω
Ld	15.48	mH
Lq	25.8	mH
PM Flux linkage	0.42	Wb
inertia	0.001469	$Kg \cdot m^2$
Viscous coefficient.	0.0003035	N.m.s

B. Simulation Result

To evaluate the validity of the proposed method, step changes from 2 Hz to 5 Hz in speed command with step load torque have been applied to the system. Two different tests have been performed.

1) The DC bus voltage level is 300 volt

Figure 4 shows the behavior of the actual and estimated values of the speed, the rotor position and torque. The actual steady state speed is equal to $\omega=31.4$ rad/s (frequency is 5Hz). It can be seen that the estimated speed follows weakly the actual speed. The high ripple at low speeds is mainly due to the voltage measurements. At low speeds, the voltage signals are smaller in magnitude and their fundamental components are comparable in magnitude to the neglected harmonics.

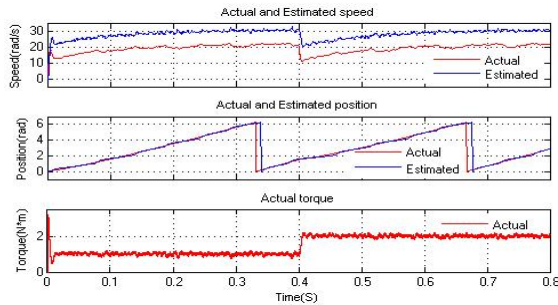


Fig 4. Transient and steady state behavior of the estimated /actual speed and position. $\theta_r(0)=0$, T_L step changed from 1 N*m to 2N*m.

Figure 5 shows the behavior of the actual and estimated values of the speed, the rotor position and torque. The actual steady state speed is equal to $\omega=12.56$ rad/s (frequency is 2Hz). It can be seen that the estimated speed follows unsuccessfully. At low

speeds, the voltage signals are smaller in magnitude and their fundamental components are disturbed by the neglected harmonics seriously. This loss of information affects the EKF performance. However, by adjusting the DC bus voltage at low speeds, the steady state errors and the ripple components can be minimized.

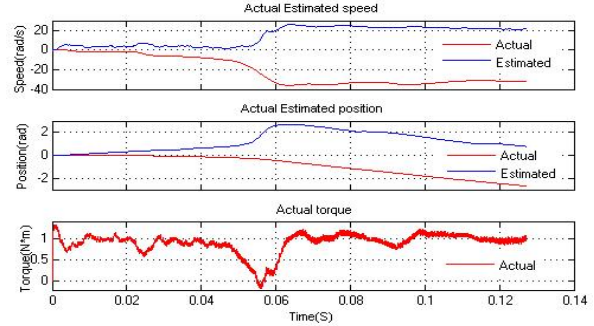


Fig 5. Transient and steady state behavior of the estimated/actual speed and position. $\theta_r(0)=0$, T_L step changed from 1 N*m to 2N*m.

2) The DC bus voltage is 30 volt

The aforesaid simulation test results have provided that the performance is not the optimal especially at low speed. It has been found that the DC bus voltage decreased appropriately will improved the performance of the EKF control system at low speed from chapter IV experimental data. The DC voltage level must be modified in order to acquire higher performance at low speed. The regulation is to change the DC bus voltage to 30 volt.

Figure 6 shows the behavior of the actual and estimated values of the speed, the rotor position and torque. The actual steady state speed is equal to $\omega=31.4$ rad/s (frequency is 5Hz). It can be seen that the estimated speed follows closely the actual speed after an initial transient. Similarly, the estimated position go through a transient and then converges to follow the actual position. The transient duration is about 100 ms. It is apparent that the system have a good performance both at transient and steady state while speed is equal to 31.4 rad/s.

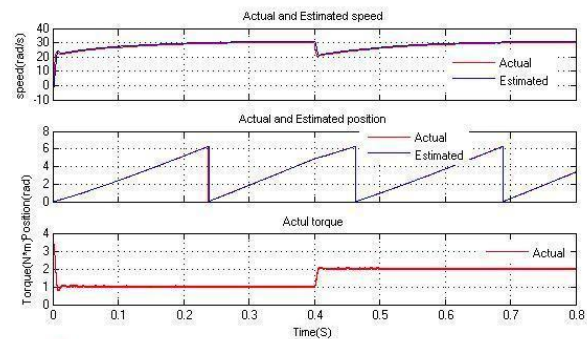


Fig 6. Transient and steady state behavior of the estimated/actual speed and position. $\theta_r(0)=0$, T_L step changed from 1 N*m to 2N*m.

Figure 7 shows the behavior of the actual and estimated values of the speed, the rotor position and torque. The actual steady state speed is equal to $\omega=12.56$ rad/s (frequency is 2Hz). It can be seen that the estimated speed follows fairly the actual speed. The speed error is mainly due to the voltage measurements and the PI regulator parameters.

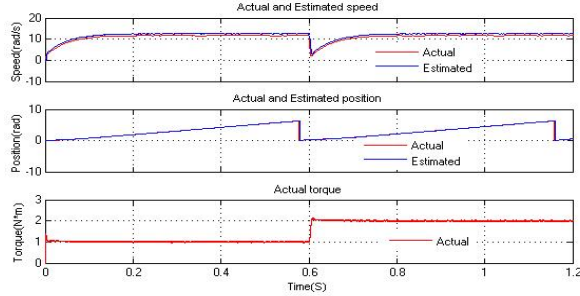


Fig 7. Transient and steady state behavior of the estimated/actual speed and position. $\theta_r(0)=0$, T_L step changed from 1 N*m to 2 N*m.

It is clear that the extended Kalman filter tracks very well the speed and rotor position of the motor at high speed and low speed with adjustable DC bus voltage. But, the performance is weakening at low speed (<1Hz). The voltage signals are smaller in magnitude and their fundamental components are disturbed by the neglected harmonics and measurement error seriously when speed is less than 1 Hz. So, the EKF can not work at zero speed. It is a disadvantage of EKF control system.

VII. EXPERIMENTAL RESULT

The experiment has been realized on DSpace 1104 system. The EKF was tested in steady state and transient operation. The parameters of the PMSM are listed in Table 1.

Figure 8,9,10 show the experimental waves, from the left to the right and top to bottom, are as follows: three-phase line voltage, estimated/actual position, speed as reference and estimated/actual speed, dc bus voltage as well as two phase voltage in the stationary reference frame.

Figure 8 presents the laboratory results when the speed was 314 rad/s and dc bus voltage was 245V. The figure shows the better performance of the speed of the system and position tracking in the high-speed scale; and the speed steps into steady state just after a short adjustment of the transient state, with transient state process being 150ms. Waveforms of voltage and current are in perfect condition.

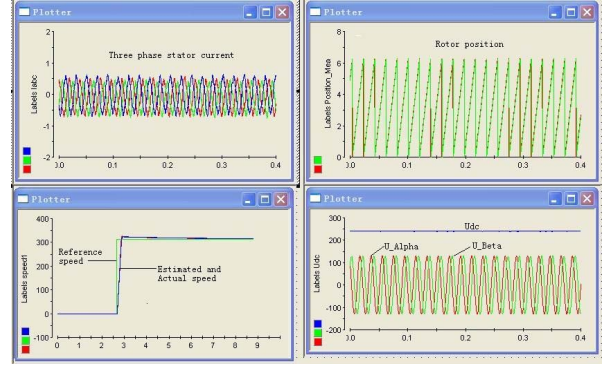


Fig 8. Test performance of the estimated/actual speed and position.

$$\theta_r(0)=0, \omega_r=314 \text{ rad/s}.$$

Figure 9 shows the experimental results when the speed was 32.4 rad/s, and the dc bus voltage was 24v. As can be seen from the figure, after the adoption of dc bus voltage drop, the system can keep stable operation around 5Hz, and the rotor position, velocity tracking perform well, dynamic response in fast condition, the duration of transient state process lasts for 50ms; voltage and current waveforms work well.

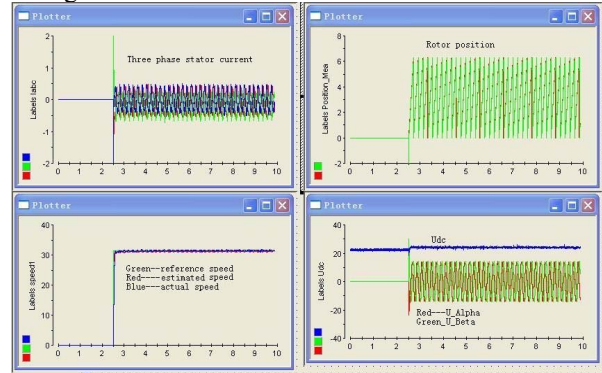


Fig 9. Test performance of the estimated/actual speed and position.

$$\theta_r(0)=0, \omega_r=31.4 \text{ rad/s}.$$

Figure 10 shows the experimental results when the speed was 12.56 rad/s, and the dc bus voltage 24V. As can be seen from the figure, by adopting the method of decreasing dc bus voltage, the system can still remain fairly good dynamic and steady performance under a low-speed circumstances of 2Hz. Estimated speed and position closely follow the actual speed and position, making better voltage waveforms, but the current waveforms act with certain distortion; when adding 1N m load around 9 seconds, we find the speed appearing in obvious fluctuation, mainly because of the drop of dc bus voltage, but not leading to failure in system tracking; actually, a smaller velocity values is reached immediately and runs in a stable state. By 16 seconds, the load is reduced to zero suddenly, and the system returns to the steady state before loading.

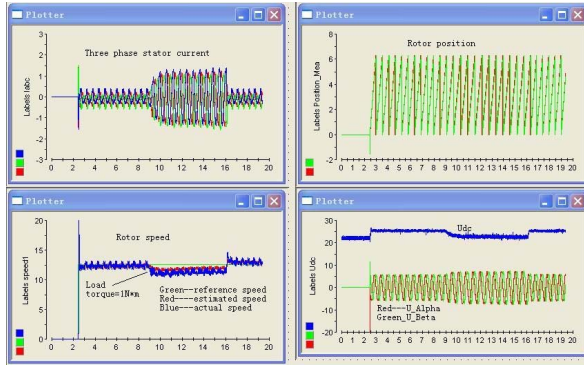


Fig 10. Test performance of the estimated/actual speed and position.

$$\theta_r(0)=0, \omega_r=12.56 \text{ rad/s}.$$

From above experimental and simulation data, we can concluded that the EKF control system have good performance both in high speed and low speed with adjustable DC bus voltage. The validity of the proposed method is provided by the test.

VIII. CONCLUSIONS

This paper puts forward a new kind of extended kalman filter method with dc bus voltage regulation law and won good performance at low velocity; Based on the analysis of kalman filtering theory, the factors affecting the accuracy -- voltage ratio of signals-noise was discovered, and the simulation experiment was made according to the proposed method, which proved the validity of the proposed method.

Based on the permanent magnet synchronous motor DSPACE 1104 experimental platform, a series of relevant experiments were made in accordance with the proposed method. Experimental data shows that by adjusting the dc bus voltage and raising the voltage ratio of signals-noise, when the system running in a low-speed less than 2Hz, good transient and steady state performance can still be reached, and such operation extends the application scope of kalman filter and this is one of the advantages of the system. Since this system is not dependent on the salience of motor, it can be applied in surface mounted permanent magnet synchronous motor.

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